



Spring Core – Interview Questions

1. What is the Spring Framework?

Cross Q1: How does Spring promote loose coupling?

Cross Q2: Can you name 5 key modules of Spring?

2. What is Dependency Injection (DI)?

Cross Q1: How does DI help in achieving loose coupling?

Cross Q2: What are the two types of DI in Spring?

3. What is Inversion of Control (IoC)?

Cross Q1: How is IoC implemented in Spring?

Cross Q2: How is IoC different from traditional control flow?

4. What is the difference between constructor and setter injection?

Cross Q1: Which one is preferred when dependency is mandatory?

Cross Q2: Can you mix both in the same bean?

5. What is the Spring Bean lifecycle?

Cross Q1: What methods are called during initialization and destruction?

Cross Q2: How can you customize lifecycle callbacks?

6. What is the role of ApplicationContext and BeanFactory?

Cross Q1: Which one is used in real-world Spring applications?

Cross Q2: How is lazy loading handled in both?

7. What are Spring annotations for defining beans?

Cross Q1: How do `@Component`, `@Service`, `@Repository` differ?

Cross Q2: Which annotation replaces `<bean>` in XML?

8. How does component scanning work in Spring?

Cross Q1: What annotation enables component scanning?

Cross Q2: How do you scan multiple packages?

9. What is the purpose of `@Autowired`?

Cross Q1: Can it be used with constructors and setters?

Cross Q2: What is the difference between `@Autowired` and `@Inject`?



10. How do you configure Spring using XML vs Java Config?

Cross Q1: What annotation is used to enable Java-based config?

Cross Q2: Which approach is recommended in modern applications?

11. What is the use of @Qualifier?

Cross Q1: What happens if there are multiple beans of the same type?

Cross Q2: Can you use @Qualifier in constructor injection?

12. What is Aspect-Oriented Programming (AOP)?

Cross Q1: How does AOP help with cross-cutting concerns?

Cross Q2: Give real examples of cross-cutting concerns.

13. What are join points and pointcuts in Spring AOP?

Cross Q1: What types of methods can be join points?

Cross Q2: How do you declare a pointcut expression?

14. What are different types of advice in Spring AOP?

Cross Q1: What is the difference between @Before and @Around?

Cross Q2: When is @AfterReturning used?

15. How is AOP implemented internally in Spring?

Cross Q1: What are proxy objects?

Cross Q2: What's the difference between JDK and CGLIB proxy?

16. What are the different scopes of Spring beans?

Cross Q1: What is the default scope of a bean?

Cross Q2: How does prototype differ from singleton scope?

17. What is the role of the @Value annotation?

Cross Q1: How can you inject values from a .properties file?

Cross Q2: Can you inject environment variables using @Value?

18. What is the @PostConstruct and @PreDestroy annotation used for?

Cross Q1: Can they replace init-method and destroy-method in XML?

Cross Q2: Do these work on prototype-scoped beans?

19. What is the difference between @ComponentScan and @Import?

Cross Q1: Which one is used to import additional configuration classes?



Cross Q2: Can both be used in the same config class?

20. What is the difference between `@Component` and `@Bean`?

Cross Q1: Which one is used for third-party class bean creation?

Cross Q2: Can both be used together?

21. What is the use of Environment in Spring?

Cross Q1: How do you read application properties dynamically?

Cross Q2: What is `@PropertySource`?

22. What is a circular dependency in Spring?

Cross Q1: How does Spring handle it with constructor injection?

Cross Q2: How to resolve circular dependency using setter injection?

23. What is a proxy object in Spring?

Cross Q1: When do you encounter proxy objects?

Cross Q2: Can you see the proxy class name in runtime?

24. What is the purpose of `@Profile` annotation?

Cross Q1: How do you activate a profile?

Cross Q2: Can you define beans conditionally using profiles?

25. How do you test Spring applications?

Cross Q1: What is `@SpringBootTest`?

Cross Q2: How do you mock dependencies in a Spring test?

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